Why Opaque Refs Beat Brittle Selectors

Anti-Staleness Verification, Form Sanitization & Deterministic Element Resolution

Deterministic Element Targeting: Eliminating Broken Clicks

In modern web portals, asynchronous JavaScript frequently alters DOM elements between agent reasoning steps. Fathom implements an active **Anti-Staleness Guard** to guarantee that actions are executed only on valid, intended targets.

1. PRE-EXECUTION FRESHNESS CHECK

Before executing a `computer_click` or `computer_type` command, the runtime takes a microsecond snapshot to confirm that the target element ref (@e14) still exists and matches its original role fingerprint.

2. STALE-REF REJECTION & SELF-RECOVERY

If the page navigated or a popup closed during agent thinking, the ref is rejected immediately with an explicit error: *"Ref @e14 is stale."* The agent captures a fresh snapshot and re-evaluates safely.

Form Data Security & Workspace Confinement

Automatic Password Scrubbing

Password inputs and token fields are automatically masked in accessibility snapshots.

Confined File Workspace

Browser downloads are strictly confined to an isolated directory (`/data/browser`).

Zero Phantom Interactions: Fathom ensures that automated browser actions are as predictable, deterministic, and safe as compiled software code.